

MARITIME COMPLIANCE TOOLS

USER GUIDE

FUELEU Maritime Calculator—Web Version



FUELEU Maritime Calculator - Compliance balance, GHG Emission

Prepared for: Vessel Compliance & Reporting Teams

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1. Overview of the Tool

The FuelEU Maritime Calculator is a single-page, browser-based worksheet that helps you work out a ship's GHG (greenhouse gas) intensity, FuelEU Maritime compliance balance, potential penalty exposure, and CII (Carbon Intensity Indicator) rating.

The tool is organized into five parts, matching the structure of the FuelEU Maritime regulation reporting template:

- Part A — Ship particulars, reporting period and operational data
- Part B — Fuel consumption by pathway, with GHG intensity inputs
- Part C — Calculated compliance balance and penalty
- Part D — Multi-year compliance outlook
- Part E — CII rating and outlook

Everything runs locally in your browser — no data is sent to a server. You can save your work as an XML file and reload it later, and you can export results to PDF or Excel.

The screenshot displays the FuelEU Maritime Calculator interface, organized into three main sections: Part A, Part B, and Part C.

Part A - Data identifying the ship and company
SHIP PARTICULARS, REPORTING PERIOD AND OPERATIONAL INPUTS

Inputs for Part A include:

- Name of ship
- IMO number
- Reporting year: 2025
- Reporting period From: mm/dd/yyyy
- Reporting period To: mm/dd/yyyy
- Total distance travelled (nm): 0
- Time spent at sea (hours): 0
- Time at anchorage (hours): 0
- Total number of fuels: 20
- Wind-assisted propulsion (applicable): NO
- Ice class: NA
- Ship type for CII: Bulk carrier
- Capacity for CII (DWT): Enter DWT

Part B - GHG intensity calculation by fuel
FUEL PATHWAYS, TONNES ALLOCATION AND EMISSION FACTOR INPUTS

Buttons: EXCEL FORMAT, IMPORT EXCEL, CLEAR

Part C - Compliance balance and penalty calculations
COMPUTED COMPLIANCE BALANCE, TARGET COMPARISON AND PENALTIES

Inputs for Part C include:

- SHOW DETAILED RESULT: NO
- SELECTED FWIND: 1.00
- REDUCTION BAND: 2025-2029 -2%
- REPORTING PERIOD SUMMARY: Select From and To dates

Key results displayed:

- Total GHG Intensity: **82.19150**
Actual = Weighted WTT + Weighted TTW, adjusted by fwind only if wind is enabled.
- Current Target: **89.3368**
Reference value 91.16 gCO₂eq/MJ reduced by FuelEU year band.
- Compliance Balance: **584.96 tCO₂eq**
Displayed in tCO₂eq.
- Status: **Compliant**
Penalty: EUR NA
- TOTAL ENERGY IN SCOPE: **81,866.60**
MJ
- WEIGHTED WTT: **18.46138**
 $\frac{\sum(WTT \times \text{energy})}{\sum(\text{energy} \times \text{reward})}$
- WEIGHTED TTW: **63.73012**
 $\frac{\sum(TTW \times \text{energy})}{\sum(\text{energy} \times \text{reward})}$
- FUELEU PENALTY: **EUR NA**
Only calculated when balance is negative

ENERGY CONSUMPTION SUMMARY

Energy consumption	EEA-EEA	EEA-non EEA	non EEA-EEA	Total
	From Voyages between ports under a MS jurisdiction	From Voyages which departed from ports under a MS jurisdiction	From Voyages to ports under a MS jurisdiction	
Fuel amount considered [m tonnes]	220.00	80.00	1,391.87	1,691.9

Tip: Because the tool works entirely in your browser, nothing is saved automatically. Use the Save XML button regularly so you don't lose your entries if you close the tab.

2. Getting Started — Opening the Calculator

1. Locate the file named similar to FuelEU_Calculator_Final.html on your computer.
2. Double-click the file — it will open in your default web browser (Chrome, Edge, or Firefox all work well).
3. No installation or internet connection is required to use the calculator itself (only the initial page load uses a couple of small online libraries for charts and Excel import/export).
4. The calculator loads with all fields empty and ready for input, or you can click Load sample case to see a pre-filled example first.

Tip: If you regularly use this tool, consider bookmarking the file or saving a shortcut to your desktop for quick access.

3. The Top Toolbar

A row of buttons sits at the top of the page. These control the whole worksheet, not just one section.

Button	What it does
Calculate now	Runs (or re-runs) the full calculation across Parts A–E using whatever data is currently entered. Use this any time you want to refresh the results immediately.
Load sample case	Clears the form and fills it with a ready-made example, so you can see how the calculator behaves before entering your own data.
Save XML	Downloads all of your current entries as an .xml file. This is the tool's built-in "save" function — see Section 9 for details.
Import XML	Loads a previously saved .xml file back into the form, restoring every field exactly as it was.
Print / Save PDF	Opens your browser's print dialog, formatted for a clean PDF export of the completed report — see Section 9.
Reset	Clears every field on the page back to blank. This cannot be undone, so save your work first if you need it.
? (Page Help)	Opens a short in-app help panel summarizing how the page works.



Important: Reset permanently clears all entered data. Always click Save XML first if you might need the data again.

4. Part A — Ship & Voyage Data

Part A captures the ship's identity, the reporting period, and the operational totals used throughout the rest of the calculator.

Fields to complete

- Name of ship and IMO number — identifying details for the vessel.
- Reporting year — select the applicable FuelEU/CII reporting year.
- Reporting period From / To — the start and end dates of the period being reported.
- Total distance travelled (nm) and Time spent at sea (hours) — totals for the reporting period.
- Time at anchorage (hours) — used in supporting calculations.
- Total number of fuels — how many separate fuel/pathway rows you expect to enter in Part B.

- Ice class — applies a default ice class to all Part B rows (can be overridden per row).
- Ship type for CII and Capacity for CII (DWT) — required later for the Part E CII rating.
- Wind-assisted propulsion (applicable) — set to YES if the ship uses wind-assist technology; this enables the fwind reduction factor in Part C.

Tip: Fill in Part A completely before moving to Part B — several Part B and Part E calculations rely on values entered here (ice class, reporting year, ship type, and capacity).

5. Part B — Fuel & GHG Intensity Inputs

Part B is where you list every fuel pathway or energy consumer used by the ship, with the tonnage and GHG-related values for each one. This section drives the GHG intensity, compliance balance and CII calculations.

Adding and removing rows

- Click + Add Row at the bottom of the table to add a new fuel/pathway line.
- Click Remove on any row to delete that entry.
- For each row, choose the Pathway name / Consumer (fuel type), enter the Fuel amount (tonnes), the Voyages/Ports scope, and the Distance (NM) for that row.

The Part B toolbar (above the table)

Button	What it does
ExcelFormat	Download a blank Excel template matching the Part B columns, so you can prepare your fuel data offline in a spreadsheet.
Import Excel	Lets you choose a completed Excel file (based on the downloaded template) and loads all rows into Part B automatically.
Clear	Removes all imported/entered Part B rows, leaving the table empty.

Tip: For ships with many fuel types or voyages, it is usually faster to fill in the Excel Format template offline and use Import Excel rather than typing each row directly into the browser.

PoS / PoC advanced values

Where certified Proof of Sustainability (PoS) or Proof of Compliance (PoC) values are available for a fuel, click Update on that row to open the advanced inputs popup. Here you can override the default Lower Calorific Value (LCV), Emission Factor (E value), the e-fuel specific factor (eu, for e-fuels only), and the CH₄ slip percentage (for LNG pathways only) with your certified figures. Leave an item set to "No change" to keep the calculator's standard default value.

fwind (wind-assisted propulsion)

If Wind-assisted propulsion was set to YES in Part A, a popup lets you select the applicable fwind reduction factor (1, 0.99, 0.97 or 0.95). This factor is then applied automatically in the Part C compliance calculation.

Part B · GHG intensity calculation by fuel
FUEL PATHWAYS, TONNES ALLOCATION AND EMISSION FACTOR INPUTS

EXCEL FORMAT IMPORT EXCEL CLEAR A

VOYAGE NO.	PATHWAY NAME / CONSUMER	POS/POC AVAILABLE	VOYAGES/ PORTS SCOPE	DISTANCE NM	FUEL AMOUNT (M TONNES)	ACTIONS
1	Ethane	NO	EEA-EEA	0	200	Remove
2	MDO MGO (Grades DMX to DMB)	NO	EEA-non EEA	0	200	Remove
3	HFO (Grades RME to RMX)	NO	non EEA-EEA	0	30	Remove
4	Ethane	NO	non EEA-EEA	0	20	Remove
5	LNG / LNG Otto (dual fuel medium speed)	NO	non EEA-EEA	0	50	Remove
6	LNG / LNG Otto (dual fuel medium speed)	NO	non EEA-EEA	0	60	Remove
7	LFO (Grades RMA to RMD)	NO	non EEA-EEA	0	80	Remove
8	LNG / LNG Otto (dual fuel slow speed)	NO	non EEA-EEA	0	58	Remove

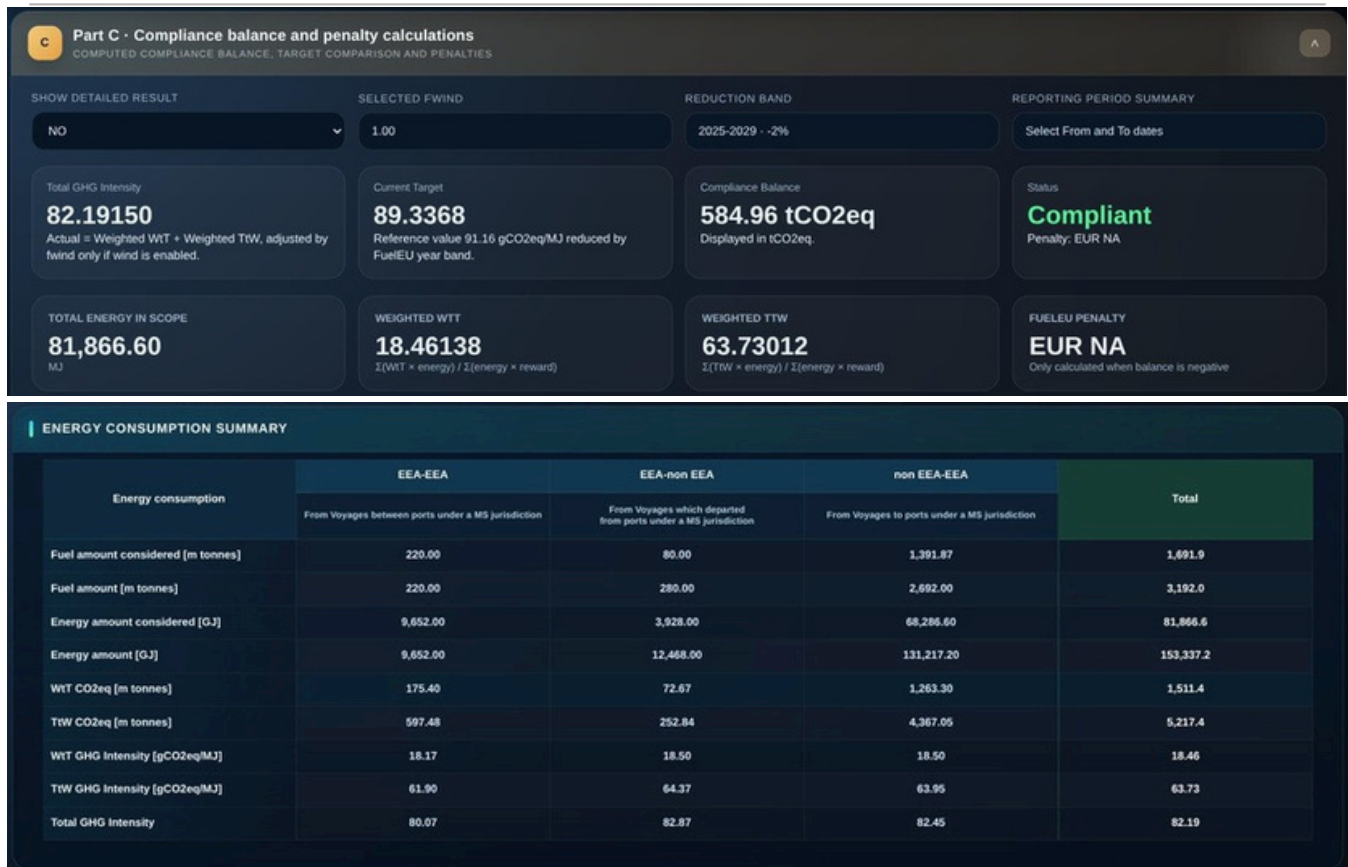
Add Row

6. Part C — Compliance Balance & Penalty

Part C is calculated automatically from Parts A and B — there is nothing to type here except the optional Show detailed result toggle.

- Total GHG Intensity (Actual) — the ship's weighted GHG intensity for the reporting period.
- Current Target — the regulatory limit for the selected reporting year.
- Compliance Balance — the surplus or deficit in tCO₂eq versus the target.
- Status and Penalty — shows whether the ship is compliant, and the estimated FuelEU penalty in EUR if the balance is negative.
- Total Energy in Scope, Weighted WtT and Weighted TtW — supporting figures used to reach the headline numbers above.

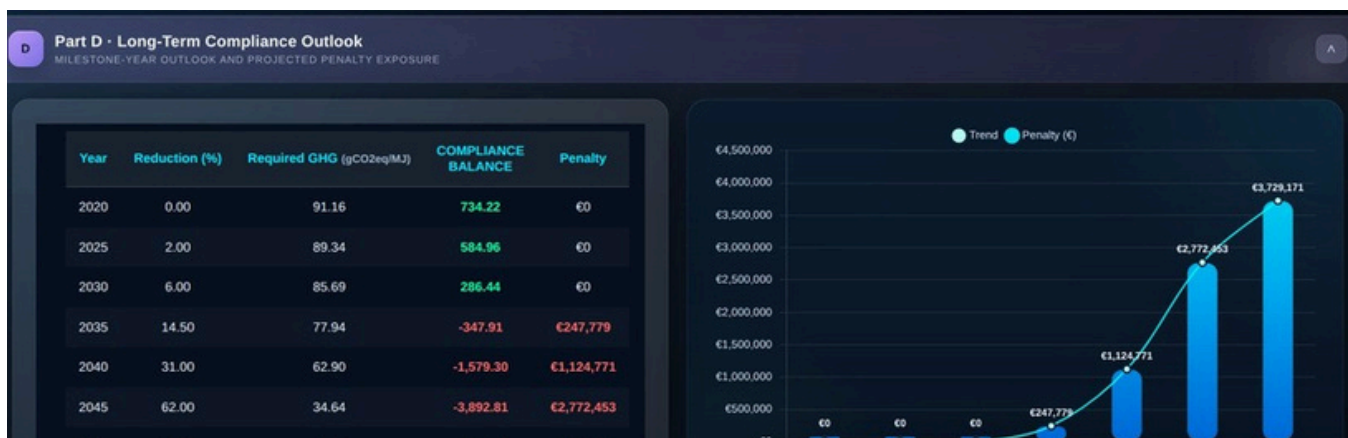
Set Show detailed result to YES to open a row-by-row breakdown showing the contribution of every Part B fuel row to the overall result.



Tip: If Part C shows all zeros, double check that Part B has at least one row with a fuel amount and distance entered, and that Calculate now has been clicked.

7. Part D — Long-Term Compliance Outlook

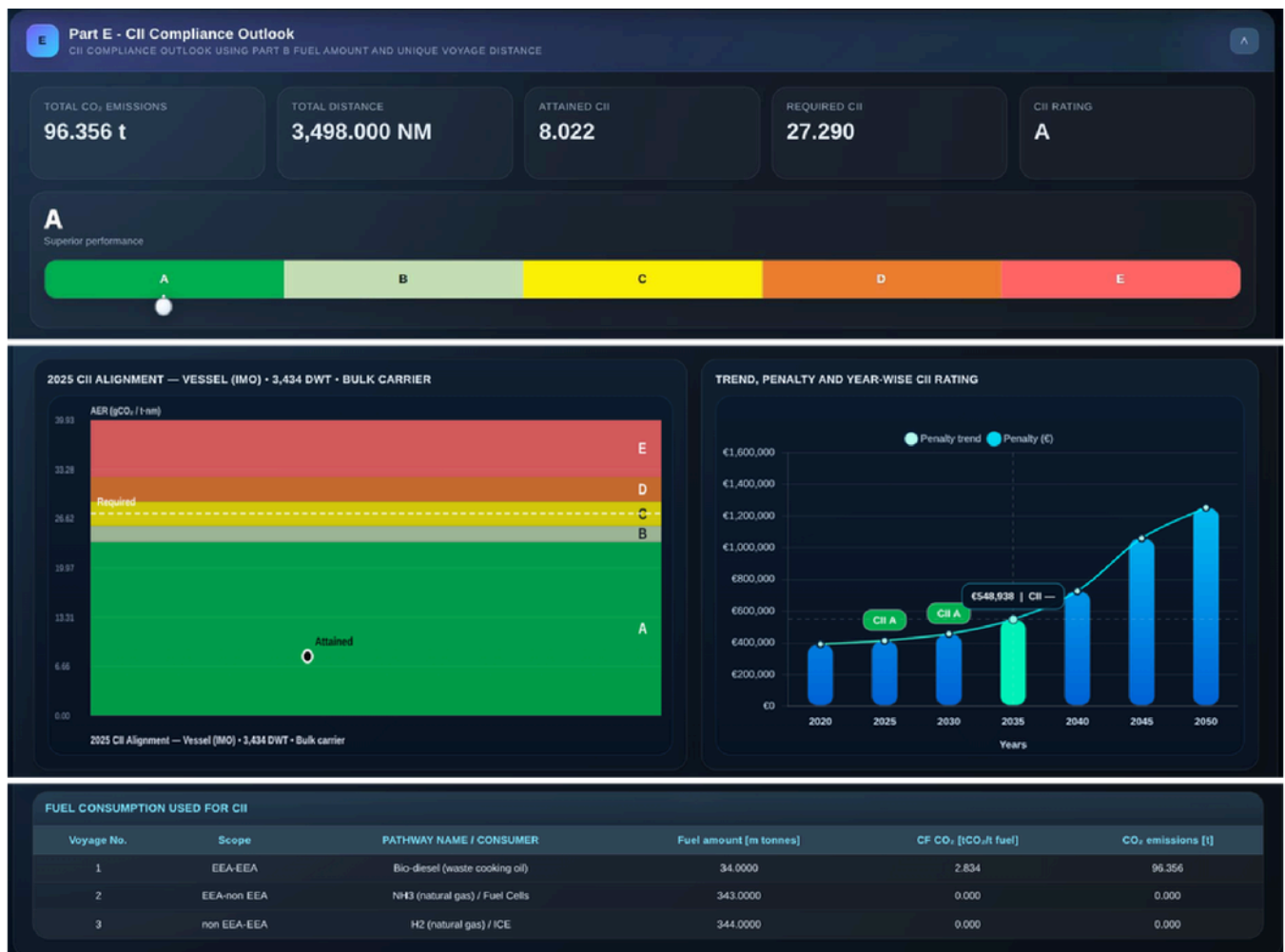
Part D projects the ship's compliance position across future FuelEU milestone years, using the same Part A and Part B data, so you can see how the compliance balance and penalty exposure are expected to change as the regulatory target tightens over time.



8. Part E — CII Compliance Outlook

Part E calculates the ship's Carbon Intensity Indicator (CII) rating using the fuel consumption entered in Part B together with the ship type, capacity (DWT/GT) and total distance entered in Part A.

- Total CO₂ emissions and Total distance — totals used in the CII formula.
- Attained CII and Required CII — the ship's calculated intensity versus the regulatory requirement for the selected year.
- CII rating — the resulting letter grade (A–E) shown against the rating scale.
- Charts — a CII graph and a multi-year trend/penalty/rating chart.
- Fuel consumption table — a per-row breakdown of fuel amount, carbon factor and CO₂ emissions used in the CII calculation.



Important: Part E will not calculate until Part B has at least one row with a fuel amount and a Distance (NM) value, and Part A has a capacity (DWT/GT) entered for the selected ship type.

9. Downloading & Saving Your Work

The calculator offers three separate ways to save or export your work, each suited to a different purpose.

9.1 Save your entries (Save XML)

5. Fill in Parts A and B as needed.
6. Click Save XML in the top toolbar.
7. Your browser downloads an .xml file (usually to your Downloads folder) containing every field you entered.
8. To resume later, open the calculator, click Import XML, and select the saved file — all fields are restored exactly as you left them.

Tip: Save XML is the file to use if you want to keep editing the same case later, share the raw data with a colleague, or keep an audit record of your inputs.

9.2 Export the Part B fuel table to Excel

9. In the Part B section, click Excel Format to download a blank spreadsheet template with the correct columns.
10. Fill in your fuel/pathway data in Excel.
11. Return to the calculator and click Import Excel, then select your completed file to load all rows at once.

Tip: This route is the fastest way to enter a long fuel list, and is also useful for keeping fuel data in a format your team can review or archive in a spreadsheet.

9.3 Save the report as a Word-readable/PDF document (Print / Save PDF)

12. Click Calculate now to make sure all figures are up to date.
13. Click Print / Save PDF in the top toolbar.
14. Your browser opens its print dialog with the report formatted for printing.
15. Under Destination (or Printer), choose Save as PDF (Windows/Chrome) or Save to PDF (Mac).
16. Choose a file name and folder, then click Save.

This produces a clean, formatted PDF snapshot of the completed calculation — ideal for sharing, filing, or attaching to a compliance record.



Tip: If you later need this report in Microsoft Word, open the saved PDF in Word (File > Open, then choose the PDF) and Word will convert it into an editable .docx document. Formatting such as charts and tables may need minor adjustment after conversion.

10. Troubleshooting & Tips

Issue	What to check
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Results show 0.00 / EUR NA everywhere	Make sure Part B has at least one row with a fuel amount, and click Calculate now.
Part E shows a warning message instead of results	Enter Distance (NM) in Part B and Capacity (DWT/GT) plus Ship type in Part A.
Import Excel does nothing	Confirm the file is based on the template downloaded via Excel Format — other layouts are not recognized.
I lost my data after closing the browser	The tool does not auto-save. Use Save XML regularly, especially before closing the tab or clicking Reset.
Print / Save PDF looks cut off or missing sections	In the print dialog, enable "Background graphics" and set margins to Default before saving.

For further help, click the ? (Page Help) icon in the top toolbar of the calculator for a quick in-app summary of these same steps.